

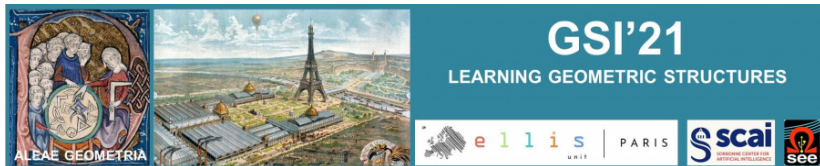
The special unitary group, monotone metrics and gradient vector fields

An intertwined tale

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1 Introduction

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Space of (mixed) Quantum States

Start from ($n \times n$ complex) matrices,

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \in \mathcal{M}_n(\mathbb{C}). \quad (1)$$

On this space the involution $\dagger$ can be defined, which has two eigenspaces,

$$\mathcal{V} = \left\{ \mathbf{a} \in \mathcal{M}_n(\mathbb{C}) : \mathbf{a} = \mathbf{a}^\dagger \right\}; \quad i\mathcal{V} = \left\{ \mathbf{b} \in \mathcal{M}_n(\mathbb{C}) : \mathbf{b} = -\mathbf{b}^\dagger \right\}. \quad (2)$$

A **quantum state** ρ is a matrix in $\mathcal{V}$ whose eigenvalues are non-negative and sum to 1,

$$\overline{\mathcal{S}}_n = \{ \rho \in \mathcal{V} : \rho \geq 0 \text{ and } \text{Tr}(\rho) = 1 \}. \quad (3)$$

$\overline{\mathcal{S}}_n$ is called the **space of quantum states**.

Stratification of the Space of Quantum States

$\overline{\mathcal{S}}_n$ is **not** a manifold, the following results hold:

Theorem

The spaces $\mathcal{S}_n^k$ of states of rank k , with $1 \leq k \leq n$ are smooth and connected manifolds in $\mathcal{V}$ of real dimension $2nk - k^2 - 1$.

Corollary

The boundary $\partial\mathcal{S}_n$ is the union of all strata with $k < n$. It is not a smooth manifold unless $n = 2$.

Moreover $\mathcal{S}_n \equiv \mathcal{S}_n^n$ is an open set.

An example: The Qubit

The qubit is the "simplest" quantum system.

$$\rho = \frac{1}{2} (\mathbb{I}_2 + x^j \sigma_j); \quad r^2 = x^j x_j; \quad \det \rho = \frac{1}{4} (1 - r^2). \quad (4)$$

Where σ_j is the j -th Pauli matrix. Then

$$\det \rho \geq 0 \implies r \leq 1. \quad \text{Bloch Ball} \quad (5)$$

We have rank one (pure) states exactly when the equality is satisfied.

Remark

$\mathcal{S}_2$ can be seen as an open ball (of radius 1) in $\mathbb{R}^3$, pure states lie on the boundary of this ball, thus $\overline{\mathcal{S}_2}$ can be seen as a closed ball (of radius 1) in $\mathbb{R}^3$.

Group Actions

On $\overline{\mathcal{S}_n}$ we define an action of $\mathrm{SL}(n, \mathbb{C})$,

$$\alpha : \overline{\mathcal{S}_n} \times \mathrm{SL}(n, \mathbb{C}) \ni (\rho, g) \mapsto \alpha_g(\rho) = \frac{g\rho g^\dagger}{\mathrm{Tr}(g\rho g^\dagger)}. \quad (6)$$

The special unitary group is

$$\mathrm{SU}(n) = \left\{ U \in \mathrm{SL}(n, \mathbb{C}) : UU^\dagger = \mathbb{I} \right\}. \quad (7)$$

If we “restrict” the action α to unitary elements of $\mathrm{SL}(n, \mathbb{C})$ we get

$$\tilde{\alpha} : \overline{\mathcal{S}_n} \times \mathrm{SU}(n) \ni (\rho, U) \mapsto \tilde{\alpha}_U(\rho) = U\rho U^\dagger. \quad (8)$$

Group Actions: infinitesimal version

The Lie algebra of $SL(n, \mathbb{C})$ is $\mathcal{M}_n^0(\mathbb{C})$, take a curve $g_t = e^{\frac{1}{2}(\mathbf{a} - i\mathbf{b})t}$ and compute

$$\Gamma_{\mathbf{ab}}(\rho) = \left. \frac{d}{dt} \alpha_{g_t}(\rho) \right|_{t=0} = [[\rho, \mathbf{b}]] + \{\rho, \mathbf{a}\} - \text{Tr}(\mathbf{a}\rho)\rho. \quad (9)$$

We have the decomposition

$$\Gamma_{\mathbf{ab}} = \mathbb{Y}_{\mathbf{a}} + \mathbb{X}_{\mathbf{b}}; \quad \mathbb{Y}_{\mathbf{a}}(\rho) = \{\rho, \mathbf{a}\} - \text{Tr}(\mathbf{a}\rho)\rho; \quad \mathbb{X}_{\mathbf{b}}(\rho) = [[\rho, \mathbf{b}]]. \quad (10)$$

The Lie algebra of $SU(n)$ is $\mathcal{V}_0$, take the curve $U_t = e^{-\frac{i}{2}\mathbf{b}t}$ and compute

$$\tilde{\Gamma}_{\mathbf{b}}(\rho) = \left. \frac{d}{dt} \tilde{\alpha}_{U_t}(\rho) \right|_{t=0} = [[\rho, \mathbf{b}]]. \quad (11)$$

We get

$$\tilde{\Gamma}_{\mathbf{b}} = \Gamma_{0\mathbf{b}} = \mathbb{X}_{\mathbf{b}}; \quad \mathbb{X}_{\mathbf{b}}(\rho) = [[\rho, \mathbf{b}]]. \quad (12)$$

Information metrics

In Quantum Information Geometry we use completely-positive trace-preserving (**CPTP**) maps¹.

There is not a unique choice for monotone metrics in the quantum case.

A general characterization for information metric has been given by Petz.

This characterization establishes a one-to-one correspondence between operator monotone functions² f satisfying $t f(t^{-1}) = f(t)$ and $f(1) = 1$ and monotone metrics G^f .

¹A linear map $\phi : \mathcal{M}_n(\mathbb{C}) \rightarrow \mathcal{M}_m(\mathbb{C})$ is CP if $\phi \otimes id_k$ from $\mathcal{M}_n(\mathbb{C}) \otimes \mathcal{M}_k(\mathbb{C})$ to $\mathcal{M}_m(\mathbb{C}) \otimes \mathcal{M}_k(\mathbb{C})$ is positive for every k .

² f is monotone w.r.t. the partial order on $\mathcal{M}_n(\mathbb{C})$ given by: $A > B$ if $A - B$ is positive definite

Information metrics and the special unitary group

Invariance under CPTP maps of information metrics implies their unitary invariance.

This in turn implies that the fundamental vector fields $\mathbb{X}_{\mathbf{b}}$ of the action $\tilde{\alpha}$ are Killing vector fields for **all** information metrics.

The special unitary group is a sort of “universal symmetry” in Quantum Information Geometry.

An observation: BH metric

The so-called Bures-Helstrom metric G_{BH} is G_f with

$$f(t) = \frac{1+t}{2}. \quad (13)$$

Given $\mathbf{a} \in \mathcal{V}$ we can define the *expectation value function*

$$e_{\mathbf{a}}(\rho) = \text{Tr}(\mathbf{a}\rho) \quad \mathbf{a} \in \mathcal{V}. \quad (14)$$

It is possible to prove that

$$G_{BH}(X, \mathbb{Y}_{\mathbf{a}}) = \mathcal{L}_X e_{\mathbf{a}} \quad \text{for every vector field } X. \quad (15)$$

The $\mathbb{Y}_{\mathbf{a}}$ are gradient vector fields of expectation value functions w.r.t. G_{BH} .

An observation: BH metric

We have that

$$\begin{array}{ccc} \text{Fund. v. fields of } \alpha & \longrightarrow & \Gamma_{\mathbf{ab}} = \mathbb{Y}_{\mathbf{a}} + \mathbb{X}_{\mathbf{b}} \longleftarrow \text{Fund. v. fields of } \tilde{\alpha} \\ & \uparrow & \\ & \text{Gr. v. fields of e. v. functions w.r.t. BH metric} & \end{array}$$

Remark

The Lie algebra of fundamental vector fields of α can be represented by the fundamental vector fields of $\tilde{\alpha}$ together with the gradient vector fields of expectation value functions (w.r.t. Bures-Helstrom metric), equipped with the standard Lie bracket.

A general question

How general is this instance?

Specifically:

- Given a monotone metric tensor G_f on the space of quantum states, can the gradient vector fields of expectation value function w.r.t. G_f , together with $\mathbb{X}_b$ give rise to a representation of a Lie algebra?
- If so, does this algebra integrate to some group action on the space of quantum states?

How does this algebra look like?

There are some general features this algebra $\mathcal{A}$ must fulfill:

- It must be isomorphic, as a vector space, to $\mathcal{V}_0 \oplus \mathcal{V}_0$.
Let $\mathbb{Y}_{\mathbf{b}}^f$ be the gradient vector field of $e_{\mathbf{b}}$ w.r.t. G^f ,

$$\Gamma_{\mathbf{ab}} = \mathbb{X}_{\mathbf{a}} + \mathbb{Y}_{\mathbf{b}}^f. \quad (16)$$

- Given that $[\mathbb{X}_{\mathbf{a}}, \mathbb{X}_{\mathbf{b}}] = \mathbb{X}_{[[\mathbf{b}, \mathbf{a}]}$ then $\mathcal{V}_0 \oplus \{0\}$ must be a Lie subalgebra of $\mathcal{A}$ isomorphic to $\mathfrak{su}(n)$.
- It can be proven that

$$[\mathbb{X}_{\mathbf{a}}, \mathbb{Y}_{\mathbf{b}}^f] = \mathbb{Y}_{[[\mathbf{b}, \mathbf{a}]]}^f. \quad (17)$$

- The unknown part are the commutator between gradient vector fields,

$$[\mathbb{Y}_{\mathbf{a}}^f, \mathbb{Y}_{\mathbf{b}}^f]. \quad (18)$$

A particular solution

In the **particular case of the qubit**, we can start from

$$G_f = \frac{1}{1-r^2} dr \otimes dr + \frac{r^2}{1+r} \left(f \left(\frac{1-r}{1+r} \right) \right)^{-1} (d\theta \otimes d\theta + \sin^2 \theta d\phi \otimes d\phi). \quad (19)$$

Choose the Pauli matrices σ_j as a basis for traceless self-adjoint matrices and direct computations of $\mathbb{Y}_{\sigma_j}^f \equiv \mathbb{Y}_j^f$ and $\mathbb{X}_{\sigma_j} \equiv \mathbb{X}_j$ brings us to

$$[\mathbb{Y}_1^f, \mathbb{Y}_2^f] = F(r) \mathbb{X}_3, \quad (20)$$

$$[\mathbb{Y}_2^f, \mathbb{Y}_3^f] = F(r) \mathbb{X}_1, \quad (21)$$

$$[\mathbb{Y}_3^f, \mathbb{Y}_1^f] = F(r) \mathbb{X}_2. \quad (22)$$

Where

$$F(r) = \frac{2}{r} \left[(r+1)f \left(\frac{1-r}{1+r} \right) + (r-1)f' \left(\frac{1-r}{1+r} \right) \right]. \quad (23)$$

A particular solution

Now, to obtain a Lie algebra, we must set $F(r) = A$ with A constant. Operating the change of variable

$$t = \frac{1-r}{1+r}, \quad (24)$$

and defining

$$h(t) = \frac{f(t)}{1-t}, \quad (25)$$

we get

$$F(t) = t h'(t) - h^2(t) = -\frac{A}{4}. \quad (26)$$

Variables can be separated to integrate this ODE.

We will deal with the cases $A > 0$, $A = 0$ and $A < 0$ separately.

Case $A > 0$

This case gives rise to the solution

$$f_A(t) = \frac{\sqrt{A}}{2}(1-t) \frac{1+t\sqrt{A}}{1-t\sqrt{A}}. \quad (27)$$

It can be checked that $\mathbb{X}_{\mathbf{b}} + \mathbb{Y}_{\mathbf{a}}^{f_A}$ is a fundamental vector field of the action of $\mathrm{SL}(n, \mathbb{C})$ given by

$$\alpha_{A,g}(\rho) = \frac{(g\rho^{\frac{1}{A}}g^\dagger)^A}{\mathrm{Tr}\left(\left(g\rho^{\frac{1}{A}}g^\dagger\right)^A\right)}. \quad (28)$$

Note that the case $A = 1$ gives rise exactly to the Bures-Helstrom case.

Case $A = 0$

This case gives rise to the solution

$$f_0(t) = \frac{t-1}{\log t}. \quad (29)$$

This is the operator monotone function giving rise to B-K-M metric,

$$G_{BKM} = \frac{1}{1-r^2} dr \otimes dr + r \log \left(\frac{1+r}{1-r} \right) (d\theta \otimes d\theta + \sin^2 \theta d\phi \otimes d\phi). \quad (30)$$

The BKM case has been already solved to give rise to the action

$$\alpha_0(U, \mathbf{a}, \rho) = \frac{e^{U \log \rho U^\dagger + \mathbf{a}}}{\text{Tr} (e^{U \log \rho U^\dagger + \mathbf{a}})}. \quad (31)$$

In this case we don't get an action of $\text{SL}(n, \mathbb{C})$. It is an action of $\text{SU}(n) \rtimes_R \mathcal{V}$ with $R_U(\mathbf{h}) = U\mathbf{h}U^\dagger$.

Case $A < 0$

In this case, the general solution is

$$f(t) = \frac{\sqrt{-A}}{2}(t-1) \tan\left(\frac{\sqrt{-A}}{2} \log t - \frac{\sqrt{-A}}{2} c\right), \quad (32)$$

where c is an integration constant.

Considerations on the regularity of this functions led to exclude this case.

The tensor G^f wouldn't be defined everywhere on $\mathcal{S}_2$.

Conclusions and outlooks

- Given a monotone metric G^f on $\mathcal{S}_n$, the vector fields $\mathbb{X}_a$ and $\mathbb{Y}_b^f$ allow in some cases to recover a group action on the space of quantum states.
- A complete description of this instance has been given in the case of the qubit.
- We also investigated the case $n > 2$ and we found all the group action found for the qubit ($n = 2$). However, we are not able to prove if these exhaust all possibilities when $n > 2$.

Thank You for Your attention!

Backup slides 1: Proof that $[\mathbb{X}_a, \mathbb{Y}_b^f] = \mathbb{Y}_{[[b,a]]}^f$

Consider

$$\begin{aligned}\mathcal{L}_{[\mathbb{X}_a, \mathbb{Y}_b^f]} e_c &= \mathcal{L}_{\mathbb{X}_a} \left(\mathcal{L}_{\mathbb{Y}_b^f} e_c \right) - \mathcal{L}_{\mathbb{Y}_b^f} \left(\mathcal{L}_{\mathbb{X}_a} e_c \right) \\ &= \mathcal{L}_{\mathbb{X}_a} \left(G^f(\mathbb{Y}_b^f, \mathbb{Y}_c^f) \right) - \mathcal{L}_{\mathbb{Y}_b^f} \left(\mathcal{L}_{\mathbb{X}_a} e_c \right). \quad (33)\end{aligned}$$

Taking into account

$$(\mathcal{L}_{\mathbb{X}_a} e_c)(\rho) = \frac{d}{dt} \left(\text{Tr}(e^{-\frac{i}{2}at} \rho e^{\frac{i}{2}at}) \right)_{t=0} = e_{[[c,a]]}(\rho), \quad (34)$$

and the unitary invariance of G^f , we get

$$\begin{aligned}\mathcal{L}_{[\mathbb{X}_a, \mathbb{Y}_b^f]} e_c &= G^f \left([\mathbb{X}_a, \mathbb{Y}_b^f], \mathbb{Y}_c^f \right) + G^f \left(\mathbb{Y}_b^f, [\mathbb{X}_a, \mathbb{Y}_c^f] \right) - \mathcal{L}_{\mathbb{Y}_b^f} (e_{[[c,a]]}) \\ &= \mathcal{L}_{[\mathbb{X}_a, \mathbb{Y}_b^f]} e_c + G^f \left(\mathbb{Y}_b^f, [\mathbb{X}_a, \mathbb{Y}_c^f] \right) - G^f \left(\mathbb{Y}_b^f, \mathbb{Y}_{[[c,a]]}^f \right). \quad (35)\end{aligned}$$

Backup slides 1: Proof that $[\mathbb{X}_{\mathbf{a}}, \mathbb{Y}_{\mathbf{b}}^f] = \mathbb{Y}_{[[\mathbf{b}, \mathbf{a}]]}^f$

So we have

$$G^f \left(\mathbb{Y}_{\mathbf{b}}^f, [\mathbb{X}_{\mathbf{a}}, \mathbb{Y}_{\mathbf{c}}^f] - \mathbb{Y}_{[[\mathbf{c}, \mathbf{a}]]}^f \right) = 0. \quad (36)$$

Allowing to conclude

$$[\mathbb{X}_{\mathbf{a}}, \mathbb{Y}_{\mathbf{b}}^f] = \mathbb{Y}_{[[\mathbf{b}, \mathbf{a}]]}^f, \quad (37)$$

Backup slides 2: Computations for the qubit case

In the qubit case we have

$$G_f = \frac{1}{1-r^2} dr \otimes dr + \frac{r^2}{1+r} \left(f \left(\frac{1-r}{1+r} \right) \right)^{-1} (d\theta \otimes d\theta + \sin^2 \theta d\phi \otimes d\phi). \quad (38)$$

For a traceless self-adjoint operator, holds $\mathbf{a} = \sum_{j=1}^3 a_j \sigma_j$, so we have

$$e_{\mathbf{a}}(\rho) = \text{Tr}(\rho \mathbf{a}) = \frac{1}{2}(a_1 r \sin \theta \cos \phi + a_2 r \sin \theta \sin \phi + a_3 r \cos \theta). \quad (39)$$

The gradient vector field $Y_{\mathbf{a}}^f$ associated with $e_{\mathbf{a}}$ by means of G_f is

$$Y_{\mathbf{a}}^f = G_f^{-1}(de_{\mathbf{a}}, \bullet) = G_f^{-1}(\bullet, de_{\mathbf{a}}). \quad (40)$$

Backup slides 2: Computations for the qubit case

Setting $e_j \equiv e_{\sigma_j}$ and $Y_j^f \equiv Y_{\sigma_j}^f$, direct computations bring us to

$$Y_1^f = (1 - r^2) \sin \theta \cos \phi \partial_r + g(r) \left(\cos \theta \cos \phi \partial_\theta - \frac{\sin \phi}{\sin \theta} \partial_\phi \right), \quad (41)$$

$$Y_2^f = (1 - r^2) \sin \theta \sin \phi \partial_r + g(r) \left(\cos \theta \sin \phi \partial_\theta + \frac{\cos \phi}{\sin \theta} \partial_\phi \right), \quad (42)$$

$$Y_3^f = (1 - r^2) \cos \theta \partial_r - g(r) \sin \theta \partial_\theta. \quad (43)$$

With:

$$g(r) = \frac{1+r}{r} f\left(\frac{1-r}{1+r}\right) \quad (44)$$

Backup slides 2: Computations for the qubit case

The fundamental vector fields of the action $\tilde{\alpha}$ read

$$X_1 = -\sin \phi \partial_\theta - \cot \theta \cos \phi \partial_\phi, \quad (45)$$

$$X_2 = \cos \phi \partial_\theta - \cot \theta \sin \phi \partial_\phi, \quad (46)$$

$$X_3 = \partial_\phi. \quad (47)$$

Where $X_j = X_{\sigma_j}$.

The commutators between the gradient vector fields can be computed,

$$[Y_1^f, Y_2^f] = F(r) X_3, \quad (48)$$

$$[Y_2^f, Y_3^f] = F(r) X_1, \quad (49)$$

$$[Y_3^f, Y_1^f] = F(r) X_2. \quad (50)$$

Where we have set

$$F(r) = (1 - r^2)g'(r) + g^2(r). \quad (51)$$

Backup slides 2: Computations for the qubit case

Now, to obtain a Lie algebra, we must set $F(r) = A$ with A a constant, obtaining

$$(1 - r^2)g'(r) + g^2(r) = A. \quad (52)$$

This is a Riccati ODE.

Operating the change of variable

$$t = \frac{1 - r}{1 + r}, \quad (53)$$

and defining

$$h(t) = \frac{f(t)}{1 - t}, \quad (54)$$

we get

$$t h'(t) - h^2(t) = -\frac{A}{4} \quad (55)$$

Backup slides 3: How does this algebra look like?

The following abstract features can be given,

$$\mathcal{A} = \mathcal{V}_0 \oplus \mathcal{V}_0, \quad (56)$$

$$\mathcal{A} \ni \mathbf{a} = (\mathbf{a}_1, \mathbf{a}_2) \quad \mathbf{a}_1, \mathbf{a}_2 \in \mathcal{V}_0. \quad (57)$$

The properties in the slide 14 allow to state

$$\begin{aligned} [\mathbf{a}, \mathbf{b}] = [(\mathbf{a}_1, \mathbf{a}_2), (\mathbf{b}_1, \mathbf{b}_2)] &= ([\mathbf{a}_1, \mathbf{b}_1]_1, \mathbf{0}) + (\mathbf{0}, [\mathbf{a}_1, \mathbf{b}_2]_1) \\ &+ (\mathbf{0}, [\mathbf{a}_2, \mathbf{b}_1]_1) + [(\mathbf{0}, \mathbf{a}_2), (\mathbf{0}, \mathbf{b}_2)]_2. \end{aligned} \quad (58)$$

Where

$$[\mathbf{a}_1, \mathbf{b}_1]_1 = [[\mathbf{a}_1, \mathbf{b}_1]] = \frac{1}{2i}(\mathbf{a}_1 \mathbf{b}_1 - \mathbf{b}_1 \mathbf{a}_1). \quad (59)$$

The bracket $[\ , \]_2$ must be determined.